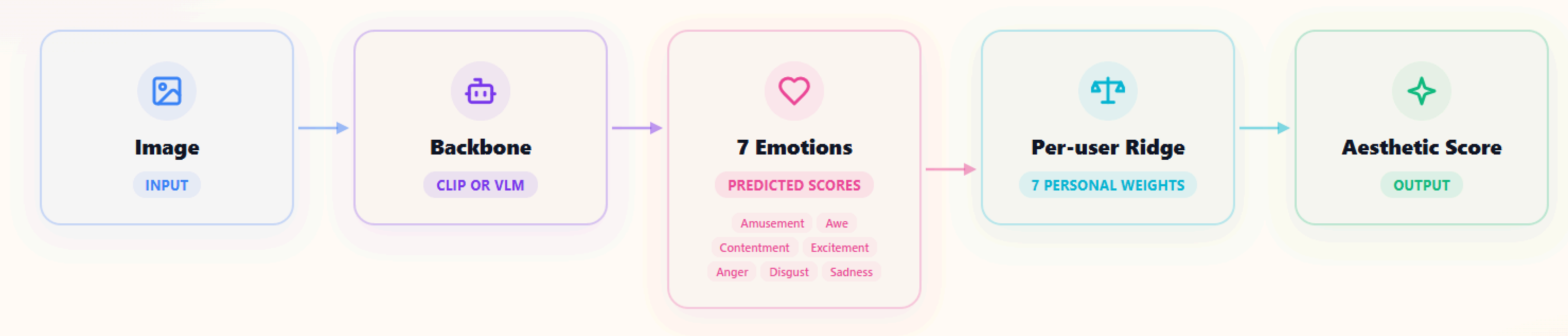


WEEKLY PROGRESS UPDATE

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses

How it works.



i The emotion layer in the middle makes the whole thing **interpretable**.

Last two meetings' Key findings.

MEETING 1 Emotion Mediation is Feasible

Feasibility & Upper Bound

We tested if using emotion as a middle step is possible. Yes: true emotions reach **0.72 (upper bound)**, beating direct beauty prediction.

The Emotion Bottleneck & Signature

We must predict emotions accurately to predict beauty well. Formulas correlate highly across domains ($p < 10^{-9}$) — showing a unique personal signature (except nostalgia in Art).

MEETING 2 Exploring the Emotion Formula

No Strong Connection to Personality

A person's emotion formula does not match their Big Five personality traits in any strong way.

Stable Across Image Types

The personal emotion weights stay highly consistent across Art, Fashion, and Landscape. Formulas correlate highly across domains ($p < 10^{-9}$) — showing a unique personal signature (except nostalgia in Art).

Three things I did.

1

A fair comparison

I re-ran Hayashi-san's ICI and MIR baselines on exactly the same data split as ours, so the comparison is fair.

2

A different backbone

I tried a vision-language model, Qwen3-VL-8B, in place of CLIP, following the Ryu & Yanaka paper Hayashi-san recommended.

3

Fine-tuning the backbone

Instead of keeping the backbone frozen, I tried fine-tuning it to predict emotions.

Making the comparison fair.

ALIGNING THE DATA SPLIT AND RESOLVING DUPLICATES TO MAKE THE BASELINE COMPARISON VALID.

BEFORE: UNALIGNED DATA SPLIT

Random split, seed=42

Had the same **COUNT** (100 train / 50 test) but used completely different **IMAGES** than ICI/MIR, making direct comparison invalid.

AFTER: ALIGNED SPLIT & DEDUPLICATION

Read train_PIAA.txt / test_PIAA.txt directly

Guarantees exact image alignment. Identified and fixed a **test-retest duplication issue** by averaging repeated ratings instead of pulling double rows.

TABLE 1 — FAIR COMPARISON, CCC BY DOMAIN (N_TRAIN=100)

Type	Model	art	fashion	landscape	avg
CEILING	P-oracle (ceiling)	0.725	0.715	0.734	0.725
BASELINE	ICI (Hiyoshi-san)	0.480	0.315	0.430	0.409
BASELINE	MIR (Hiyoshi-san)	0.484	0.313	0.435	0.411
OURS	Ours: CLIP Hybrid	0.456	0.264	0.377	0.366
OURS	Ours: CLIP Direct	0.384	0.202	0.292	0.293

Does emotion mediation really help?

WILCOXON SIGNED-RANK TEST EVALUATES WHETHER ROUTING PREDICTIONS THROUGH EMOTIONS SIGNIFICANTLY IMPROVES CCC.

TABLE 4 — WILCOXON SIGNED-RANK TEST RESULTS (COMPARING HYBRID VS DIRECT)

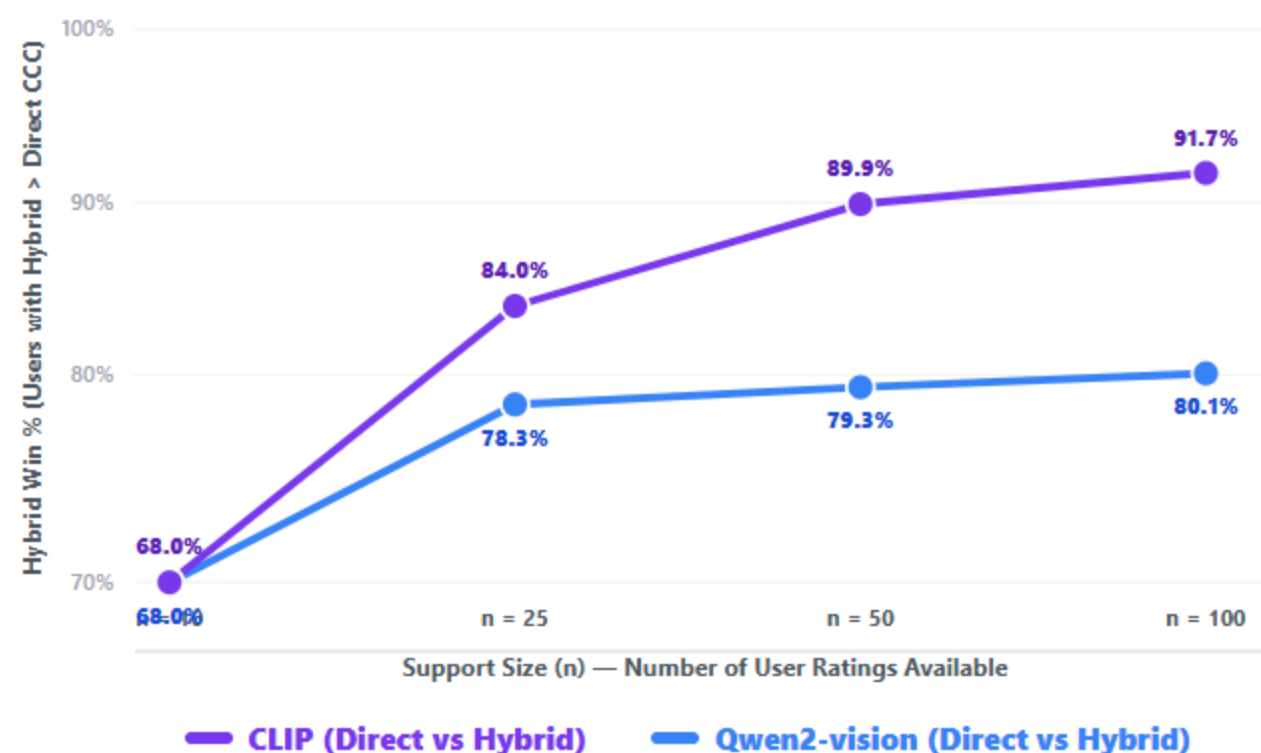
• CLIP Backbone (Direct vs Hybrid)				• Qwen2-vision Backbone (Direct vs Hybrid)			
N	MEDIAN Δ	WIN %	EFFECT R	N	MEDIAN Δ	WIN %	EFFECT R
10	+0.007	68.0%	0.36	10	+0.001	68.0%	0.38
25	+0.057	84.0%	0.70	25	+0.053	78.3%	0.62
50	+0.067	89.9%	0.79	50	+0.058	79.3%	0.64
100	+0.071	91.7%	0.82	100	+0.049	80.1%	0.62

INTERPRETATION & ANALYSIS

- **n** is the number of personalization images per user (support size).
- **Median Δ** is the median of Hybrid minus Direct CCC score per user. Higher positive value means emotion mediation helps more (e.g. **+0.071 CCC** for CLIP at n=100).
- **Win %** is the percentage of users who see positive gains (Hybrid CCC > Direct CCC). At n=100, it is **91.7%**, showing almost everyone benefits.
- **Effect r** is the Wilcoxon effect size showing how strong the difference is (0.3 = small, 0.5 = large). At n=100, we get **0.82**, which is extremely large.
- All results are statistically significant (Wilcoxon signed-rank test, **p < 0.001**).

Win rate by support size (n).

HOW THE PERCENTAGE OF USERS WITH POSITIVE GAINS (HYBRID CCC > DIRECT CCC) SCALES AS MORE USER RATINGS ARE AVAILABLE.



Data Threshold: $n \geq 25$

At $n=10$ (very few images), the emotion step barely helps (win rate 68%, effect size 0.36). But from $n=25$ and above, the numbers jump up and stay high (win rate >84%, effect size >0.6).

Robust Across Backbones

Both CLIP and Qwen2-vision show the exact same pattern. This means the result is robust across different backbones and is not a fluke of a single model.

Trying a VLM backbone: Qwen3-VL-8B.

TABLE 5 — SIDE-BY-SIDE COMPARISON WITH RYU & YANAKA

Aspect	Reference Paper	Our Approach	Status
Model	Qwen3-VL 2B / 4B / 8B	8B (Largest)	CLOSE
Layer	LT ₁₅	Stored 36 layers (L15 ≈ L32)	CLOSE
Token	Text tokens	Same (wins from mid-layer up)	MATCH
Pooling	Average	Average	MATCH
Prompt	"Assess the aesthetics..."	Same words	MATCH
Downstream	Ridge + StandardScaler	Same	MATCH
Fine-tune	None (frozen, per Fig. 1)	We tried fine-tuning (Task 3)	EXTENDED
Metric	Spearman ρ	CCC (primary) + SROCC	EXTENDED

OUR IMPLEMENTATION DETAILS

- We extracted features from all layers in a single run. We stored both **text** ("LT") and **image** ("LV") tokens from all 36 layers, then tested them to choose the best layer.
- We followed the paper exactly for feature extraction (model, layers, tokens, prompt, and scaling). We only made two changes: we used the larger 8B model instead of 4B, and we went further by fine-tuning the backbone (which the paper did not do).

Why StandardScaler matters.

Backbone	Preprocessing	CCC ↑	SROCC ↑
Qwen3-VL-8B (4096d)	L2-norm	0.006	0.385
Qwen3-VL-8B (4096d)	StandardScaler	0.376	0.413
CLIP (512d)	L2-norm	0.301	0.413
CLIP (512d)	StandardScaler	0.293	0.399

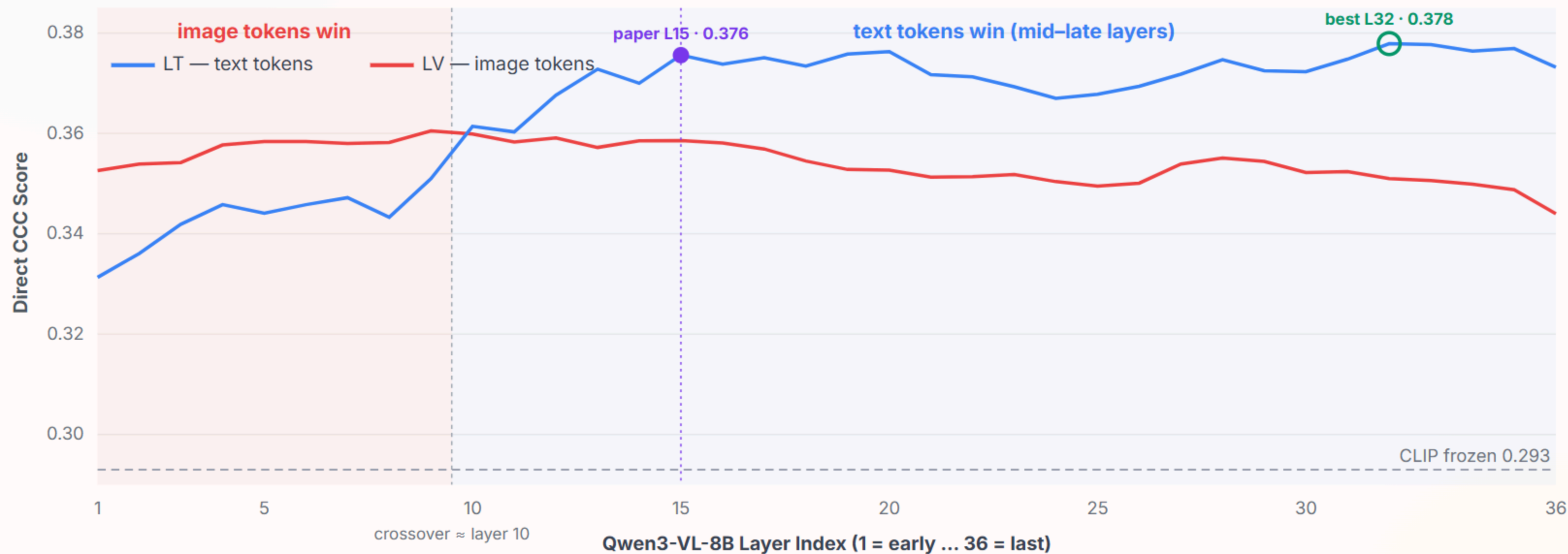
HOW STANDARDSCALER SAVES HIGH-DIM FEATURES

- **L2-norm shrinks values:** L2-normalizing 4096 dimensions makes individual values extremely tiny (near 0).
- **Ridge weight penalty:** Under tiny values, the Ridge weight decay (regularization) penalizes weights heavily, forcing predictions to collapse to the average. This destroys CCC (variance score) while SROCC (ranking) survives.
- **StandardScaler preserves variance:** Scaling each dimension individually to unit variance ($\mu=0$, $\sigma=1$) allows Ridge to assign appropriate weights.

KEY FINDINGS & ALIGNMENT

- **CLIP vs VLM:** CLIP (512d) is barely affected. The issue is strictly a high-dimensionality side effect.
- **VLM Outperforms CLIP:** With StandardScaler, the VLM beats CLIP by +0.08 CCC.
- **Text > Image Tokens:** Text tokens consistently beat image tokens in middle-to-late layers, matching the Ryu & Yanaka findings.

Which layer, and **text vs image tokens.**



As the model reads deeper, the image information moves from the image tokens into the text tokens. So the best features are text tokens at a middle-to-late layer. Layer 15 (the paper's choice) is within 0.002 of my overall best — effectively tied.

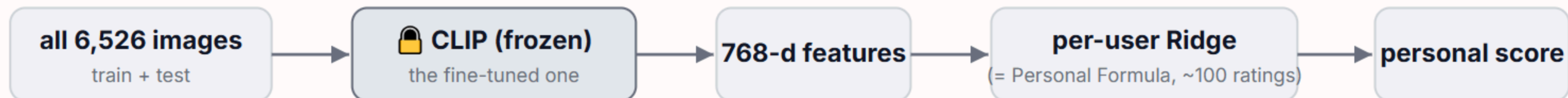
Fine-tune, then freeze, then personalize.

Stage 1 — Fine-tune (population, per fold)



↓ training done → FREEZE the fine-tuned CLIP (weights never change again)

Stage 2 — Freeze & Personalize (per user, no backprop into CLIP)



Same idea as the baselines: pretrain the backbone on the population, then personalize on top — not one joint model.

Why two stages, not one joint model? Each user has only ~100 images — fine-tuning CLIP per user would overfit. Keeping a fixed backbone plus a small linear per-user formula stays data-efficient and interpretable.

Fine-tuning & Main Comparison.

Backbone	Direct CCC	Hybrid CCC	How much the emotion step helps
CLIP frozen	0.293	0.366	+0.073
★ CLIP-ft (emotion)	0.386	0.400	+0.014
CLIP-ft (overall)	0.368	0.387	+0.019
Qwen2-vision	0.321	0.357	+0.036
Qwen3-VL-8B	0.376	0.382	+0.006
ICI (baseline)	—	0.409	—
MIR (baseline)	—	0.411	—
P-oracle (ceiling)	—	0.725	—

A small fine-tuned **CLIP (0.400)** does better than the much larger frozen Qwen3 (0.382). We don't need a huge model to get good results.

A pattern worth noting: the stronger the backbone, the less the emotion step adds (+0.073 → +0.006). A good backbone already captures emotion on its own.

These numbers are leak-free: the people we test on are never used to build the training target.

Key findings **this round.**

01 The comparison is now fair

Same data split, and duplicate ratings averaged. We now compare with ICI and MIR on exactly the same images.

02 The emotion step really helps

The Wilcoxon test shows a significant improvement for both backbones ($p < 0.001$), so it isn't down to chance.

03 It helps once you have 25+ images

With fewer than 25 images per person, the emotion step barely helps. Above that, it clearly does.

04 Scaling matters for big features

The 4096-number VLM features collapse without StandardScaler. The 512-number CLIP features are barely affected.

05 A small tuned model can win

Fine-tuned CLIP (0.400) does better than the frozen Qwen3-VL-8B (0.382). We don't need a huge model.

06 Better backbone, less to add

The stronger the backbone, the less the emotion step adds, because a good backbone already carries emotion.

What comes next.

What I plan to try next

1. Cold-start analysis

How many ratings needed before personal model beats population formula?

2. Noise-ceiling analysis

~4,500 test-retest cases. Estimate human self-consistency → true upper bound for any model.

3. Trait-conditioned emotion prediction

Can trait vectors predict how a user's emotions deviate from population? If yes → zero-shot personalization.

Questions I'd like to discuss

1. Is it worth fine-tuning Qwen too, or is "small fine-tuned CLIP beats giant frozen Qwen" strong enough?
2. Should I keep chasing accuracy, or start focusing on interpretability findings?
3. Can I use my reproduced ICI/MIR numbers (0.409/0.411) as the official comparison, since they're on exactly the same split?
4. What tier of publication should we target, and is there additional related work to study for novelty?

Not in a hurry — can continue from Thailand after the internship if needed.

Thank You!

Happy to take any questions or suggestions.

Same user, same image, **rated twice.**

The dataset intentionally re-shows a few images to the same user, to check how consistent people are with their own ratings — a test-retest design. About **5% of ratings** are these repeats, across all 129 users.

BEFORE FIX Naively selecting a user's 100 training images

1 2 3 4 5 6 7 8 9 10 11 12 13 14 ... **106** rows loaded, not 100

Two of the images (outlined in red) each appear **twice** — the duplicate rating rows get pulled in alongside the originals.

AFTER FIX Average the two ratings first, then pick 100 unique images

1 2 3 4 5 6 7 8 9 10 11 12 13 14 ... **100** exactly, every user

Hayashi-san's code

Keeps the **first** rating, drops the repeat.

My fix

Averages the two ratings — uses all the data instead of discarding it.

measured impact on CCC

+0.004

smaller than fold-to-fold noise

Why does L2-norm break a 4096-d vector?

L2-NORMALIZATION

raw feature vector (illustrative)



↓ ÷ vector length

after L2-norm — each value $\approx 1/\sqrt{4096} \approx 0.016$



Every number becomes tiny. Ridge regression can't find a useful pattern, so it plays it safe and predicts **close to the average score** for everyone.

Ranking (SROCC) survives — relative order is preserved.
Calibration (CCC) collapses — predicted values barely vary.

STANDARDSCALER

raw feature vector (same one)



↓ per-dimension $(x - \text{mean})/\text{std}$

after StandardScaler — every dimension has unit variance



Each of the 4096 dimensions is rescaled on its **own** spread, no matter how many dimensions there are. Ridge sees normal-sized numbers and fits properly.

Both ranking AND calibration recover: CCC 0.006 → 0.376.

Why CLIP (512-d) barely changes: $1/\sqrt{512} \approx 0.044$ is small but still workable for Ridge. At 4096 dimensions, $1/\sqrt{4096} \approx 0.016$ is four times smaller — that's the dimensionality tipping point where L2-norm stops working.

Ryu & Yanaka (2026) Figure 1.

THE REFERENCE PIPELINE FOR PIAA USING VLM REPRESENTATIONS: VLM HIDDEN STATES LINEARLY TRANSFORMED PER USER — NO FINE-TUNING.

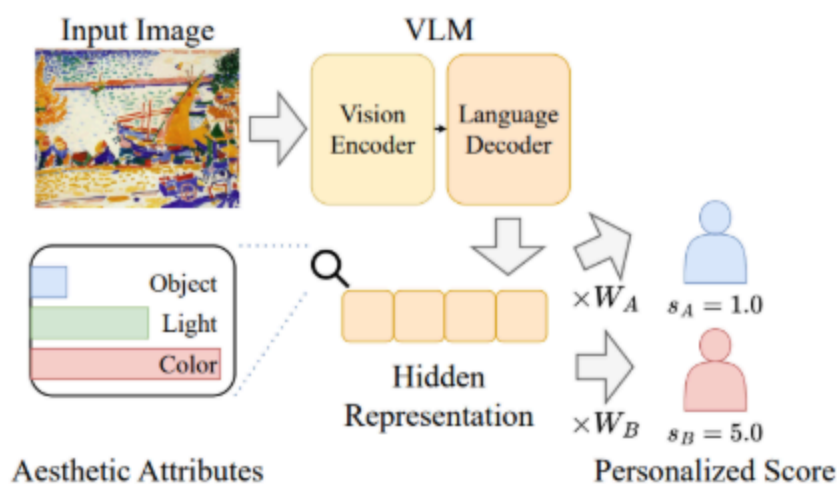


Figure 1: Overview of PIAA using VLM representations. s and W denote user-specific aesthetics scores and linear estimators respectively. Aesthetic attributes encoded in VLM hidden representations are linearly transformed to predict user-specific aesthetic scores without model fine-tuning.

Figure 1 from Ryu & Yanaka (2026) — demonstrating feature extraction without model fine-tuning.